

JANUARY 09 2023

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JASA Express Lett. 3, 014801 (2023)

<https://doi.org/10.1121/10.0016837>



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Robust underwater direction-of-arrival tracking based on variational Bayesian extended Kalman filter

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Abstract: The variational Bayesian extended Kalman filter (VB-EKF) based robust direction-of-arrival (DOA) tracking technique is proposed to make reliable estimations of the bearing angle of an underwater target with uncertain environment noise. By utilizing the VB-EKF scheme, the uncertain measurement noise caused by an unknown underwater environment along with the bearing angle of the target can be estimated simultaneously to provide reliable results at every tracking step. The proposed technique is demonstrated and verified by the sea trial data from the South China Sea in July 2021 and both the robustness and accuracy are proved superior to the traditional DOA estimating methods. © 2023 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (<http://creativecommons.org/licenses/by/4.0/>).

[Editor: John R. Buck]

<https://doi.org/10.1121/10.0016837>

Received: 22 September 2022 Accepted: 21 December 2022 Published Online: 9 January 2023

1. Introduction

Estimating the bearing angle of an uncooperative underwater target is an important research area in the field of underwater acoustic engineering.^{1–3} Considering the unknown underwater environment sometimes causes uncertain measurement noise that will degrade the estimation accuracy of the interested target's bearing angle, a robust bearing angle estimating technique is necessary and has attracted several researchers' attention.^{4,5} In addition, the passive bearing angle estimation techniques can be divided into two categories, namely, the direction-of-arrival (DOA) estimation that merely depends on the measurements from a certain period of time and the DOA tracking that depends on both the target's kinematics and the measurements of the current tracking step. Although the DOA estimation has already made a lot of milestones in various bearing angle estimation missions,^{6,7} its ignorance of the target's kinematics makes the DOA estimation techniques always have superior estimating results when the target is nearly static and the environment is stable. When the target is maneuvering or the measurement noise shifts since the underwater environment is uncertain, the DOA estimation methods can hardly have satisfactory results. Also, in order to make reliable and accurate estimations of an uncooperative underwater target's bearing angle, the DOA estimation techniques usually need a wide observation window, which will cost more on-board computational time than the recursively DOA tracking techniques.

Considering the advantages and drawbacks of the DOA estimation methods, DOA tracking techniques have been proposed. Instead of merely utilizing the measurement information, the DOA tracking methods also consider the target's kinematic model as prior information. Based on the Bayesian estimation principle, the DOA tracking techniques can reach higher estimating accuracy and occupy less on-board computing or storage resources especially when the underwater target is maneuvering.^{8,9} Also, the unknown underwater environment sometimes can make the measurement uncertain, which will degrade the bearing angle estimation accuracy or even make the estimating procedure diverge. Under this circumstance, the robust DOA tracking technique is necessary to guarantee the accuracy and robustness of the whole estimation process. Among all the robust DOA tracking techniques, two categories can be divided depending on the basic mathematical principles, namely, the "data-inspiring" robust tracking techniques¹⁰ and the Bayesian inference techniques.^{11,12} As a typical example of the "data-inspiring" robust tracking technique, the Sage-Husa adaptive estimating method (i.e., Sage-Husa extended Kalman filter, SH-EKF) has been widely used in a number of studies to prove its efficiency.^{13–15} The basic concept of the Sage-Husa online estimator is to utilize the good estimations from a long tracking period to adjust the weight of the bad measurements when the measurement noise becomes uncertain. As a result, the weights of the poor measurements can become very small to make the current bearing angle estimations mainly depend on the historically good estimations. Nevertheless, similar to many of the data-driven methods, the Sage-Husa online estimator cannot guarantee its convergence and depends heavily on good historical estimations and measurements. When the

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hyper-parameters of the data-inspiring robust DOA tracking technique (i.e., SH-EKF) are set inappropriately, the estimations usually have poor accuracy and sometimes diverge.¹³

To overcome the drawbacks of the data-inspiring robust tracking techniques, the Bayesian inference techniques, or the so-called variational Bayesian (VB) robust tracking techniques, are proposed and have been utilized in various research.^{16–20} Unlike the data-inspiring robust tracking methods, the VB robust tracking techniques are strictly derived mathematically. By assuming the prior probability density functions (PDFs) of the tracking parameters and the interested time-varying parameters (i.e., covariance matrix of the uncertain measurements noise) along with optimizing the Kullback-Leibler Divergence (KLD) among the posterior PDFs, the VB technique can analytically estimate the tracking parameters and other interesting parameters simultaneously. Different from the data-inspiring robust tracking methods, the VB robust tracking techniques have strictly mathematical foundations and can guarantee convergence and accuracy if the prior PDFs are set properly. Under this circumstance, the VB robust tracking techniques do not need good estimations from a long period to output reliable results when measurements become uncertain and are more theoretically sophisticated.

Based on the previous analysis, a variational Bayesian based extended Kalman filter (VB-EKF) is proposed to make robust and accurate DOA tracking of an underwater target with uncertain measurement noise. First, a uniform circular array (UCA) that is not only free from the port and starboard ambiguity problem but also possesses a uniform aperture at all bearing angles is considered to provide measurements of the underwater target.¹³ Then, considering the unknown underwater disturbance can sometimes make the measurement noise uncertain, the VB-EKF is designed to simultaneously estimate the bearing angle of the underwater target and the covariance matrix of the uncertain measurement noise. Finally, based on the sea trial data from the South China Sea in July 2021, the proposed VB-EKF is verified. The robust and accurate estimation results proved the superior characteristics of the proposed DOA tracking method.

2. Problem formulation

2.1 Kinematic model of the underwater target by bearing angle

Since the underwater target is usually performing in a constant velocity to keep invisibility and is far from the hydrophone arrays, the kinematic model of the target's bearing angle can be modeled as the constant velocity (CV) model.

Let θ_k denotes the bearing angle of the target (θ_k is the angle between the target and the positive x -axis with the positive counterclockwise direction) and $\dot{\theta}_k$ denotes the change rate of θ_k . Then, the CV motion kinematic model of the bearing angle is expressed as

$$\begin{cases} \mathbf{x}_k = \mathbf{F}_{k|k-1}\mathbf{x}_{k-1} + \mathbf{G}_{k-1}\mathbf{w}_{k-1}, \\ \mathbf{F}_{k|k-1} = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix}, \mathbf{G}_{k-1} = \begin{bmatrix} T^2/2 \\ T \end{bmatrix}, \end{cases} \quad (1)$$

where $\mathbf{x}_k = (\theta_k, \dot{\theta}_k)^T$ is the system state at time step k , and $(\cdot)^T$ denotes the matrix transposition, $\mathbf{F}_{k|k-1}$ and $\mathbf{G}_k$ are the state transition matrix for CV motion and the noise driving matrix, and $\mathbf{w}_{k-1}$ denotes the zero mean Gaussian process noise with the covariance matrix $\mathbf{Q}_{k-1}$ caused by the unknown underwater environment, T is the interval between the adjacent time steps. $\mathbf{Q}_{k-1}$ is named process noise covariance matrix (PNCM) for convenience.

2.2 Measurement model based on the received signal of UCA

A UCA with P elements and radius of r is used to obtain the measurement in this section. The configuration of the UCA and the underwater target is shown in Fig. 1.

The measurement of the UCA at time step k can be expressed as¹³

$$\mathbf{z}_k = \mathbf{h}_k(\mathbf{x}_k) + \mathbf{v}_k = \text{real}(\mathbf{A}(\theta_k)s(k)) + \mathbf{v}_k, \quad (2)$$

where $\mathbf{h}_k(\mathbf{x}_k)$ denotes the nonlinear measurement function, $\text{real}(\cdot)$ denotes the real part, $\mathbf{A}(\theta_k)$ is the array manifold vector of the UCA, and $s(k)$ is the narrowband acoustic signal emitted by the target, $\mathbf{v}_k$ is the measurement noise vector.

The signal of the target $s(k)$ is given by

$$s(k) = a \exp(j(2\pi f k T + \phi f_0)), \quad (3)$$

where f , a , ϕf_0 denote the frequency, amplitude, and initial phase of the signal, respectively.

The array manifold vector of the UCA $\mathbf{A}(\theta_k)$ is given as follows:

$$\begin{cases} \mathbf{A}(\theta_k) = [A_1(\theta_k), A_2(\theta_k), \dots, A_P(\theta_k)]^T, \\ A_p(\theta_k) = \exp(j2\pi f r / c \cos(\theta_k - 2\pi p / P)), \end{cases} \quad (4)$$

where p is the element number and $p = 1, 2, \dots, P$.

Since the measurement noise of each element is always dependent on one another, the measurement noise $\mathbf{v}_k$ at time step k is defined as a zero mean Gaussian noise with P -order off diagonal covariance matrix $\mathbf{R}_k$. In addition, the

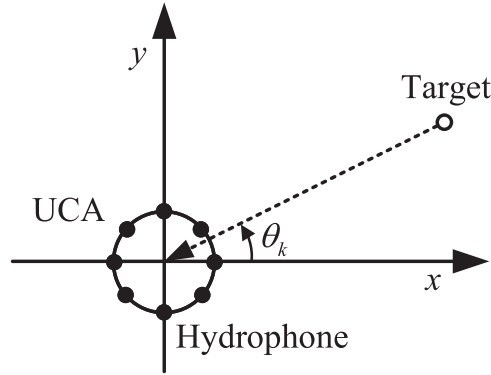


Fig. 1. Configuration of the UCA based measurement system.

measurement noise covariance matrix (MNCM) $\mathbf{R}_k$ is usually assumed to be fixed in the underwater tracking problem. However, in the real underwater tracking scenario, the unknown underwater environment can result in uncertain measurement noise, which makes the MNCM fluctuant and hard to deal with when using traditional tracking techniques. This problem seriously degrades tracking performance.

3. VB-EKF for robust DOA tracking

Noticing the measurement model with high nonlinearity depicted by Eq. (2), the extended Kalman filter (EKF) technique is used for DOA tracking in this section. Furthermore, considering the fluctuations of the MNCM caused by the unknown underwater environment, the VB-EKF¹² is utilized to improve the tracking performance by estimating the MNCM. Thus, the pseudo-code of VB-EKF for DOA tracking using the UCA at one time step is given in Algorithm 1. In order to improve the performance of MNCM estimation, some improvements are made to the algorithm.

Algorithm 1: One time step of the VB-EKF-based DOA tracking using the UCA.

Inputs: $\hat{\mathbf{x}}_{k-1|k-1}$, $\mathbf{P}_{k-1|k-1}$, $\mathbf{z}_k$, $\hat{\mathbf{u}}_{k-1|k-1}$, $\hat{\mathbf{U}}_{k-1|k-1}$, $\mathbf{Q}_{k-1}$, ρ , N .

Time update

1. $\hat{\mathbf{x}}_{k|k-1} = \mathbf{F}_{k|k-1} \hat{\mathbf{x}}_{k-1|k-1}$.
2. $\mathbf{P}_{k|k-1} = \mathbf{F}_{k|k-1} \mathbf{P}_{k-1|k-1} \mathbf{F}_{k|k-1}^T + \mathbf{G}_{k-1} \mathbf{Q}_{k-1} \mathbf{G}_{k-1}^T$.
3. $\hat{\mathbf{u}}_{k|k-1} = \rho(\hat{\mathbf{u}}_{k-1|k-1} - n - 1) + n + 1$, $\hat{\mathbf{U}}_{k|k-1} = \rho \hat{\mathbf{U}}_{k-1|k-1}$.

Iteration measurement update

4. Initialization: $\hat{\mathbf{x}}_{k|k}^{(0)} = \hat{\mathbf{x}}_{k|k-1}$, $\hat{\mathbf{R}}_k^{(0)} = \hat{\mathbf{U}}_{k|k-1} / \hat{\mathbf{u}}_{k|k-1}$.

For $i = 0 : N - 1$

Update $q^{(i+1)}(\mathbf{x}_k) = \mathcal{N}(\mathbf{x}_k; \hat{\mathbf{x}}_{k|k}^{(i+1)}, \mathbf{P}_{k|k}^{(i+1)})$

5. $\mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i)})$ is calculated by Eq. (2), $\mathbf{H}_k^{(i)}$ is calculated by Eq. (6).

6. $\mathbf{K}_k^{(i+1)} = \mathbf{P}_{k|k-1} (\mathbf{H}_k^{(i)})^T (\mathbf{H}_k^{(i)} \mathbf{P}_{k|k-1} (\mathbf{H}_k^{(i)})^T + \hat{\mathbf{R}}_k^{(i)})^{-1}$,

7. $\hat{\mathbf{x}}_{k|k}^{(i+1)} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k^{(i+1)} (\mathbf{z}_k - \mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i)}) - \mathbf{H}_k^{(i)} (\hat{\mathbf{x}}_{k|k-1} - \hat{\mathbf{x}}_{k|k}^{(i)}))$,

8. $\mathbf{P}_{k|k}^{(i+1)} = \mathbf{P}_{k|k-1} - \mathbf{K}_k^{(i+1)} \mathbf{H}_k^{(i)} \mathbf{P}_{k|k-1}$.

Update $q^{(i+1)}(\mathbf{R}_k) = \text{IW}(\mathbf{R}_k; \hat{\mathbf{u}}_{k|k}^{(i+1)}, \hat{\mathbf{U}}_{k|k}^{(i+1)})$

9. $\mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i+1)})$ is calculated by Eq. (2), $\mathbf{H}_k^{(i+1)}$ is calculated by Eq. (6).

10. $\mathbf{B}_k^{(i+1)} = (\mathbf{z}_k - \mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i+1)}))(\mathbf{z}_k - \mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i+1)}))^T + \mathbf{H}_k^{(i+1)} \mathbf{P}_{k|k}^{(i+1)} (\mathbf{H}_k^{(i+1)})^T$,

11. $\hat{\mathbf{U}}_{k|k}^{(i+1)} = \hat{\mathbf{U}}_{k|k-1} + \mathbf{B}_k^{(i+1)}$, $\hat{\mathbf{u}}_{k|k}^{(i+1)} = \hat{\mathbf{u}}_{k|k-1} + 1$,

12. $\hat{\mathbf{R}}_k^{(i+1)} = \hat{\mathbf{U}}_{k|k}^{(i+1)} / \hat{\mathbf{u}}_{k|k}^{(i+1)}$.

End for

13. $\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k}^{(N)}$, $\mathbf{P}_{k|k} = \mathbf{P}_{k|k}^{(N)}$, $\hat{\mathbf{u}}_{k|k} = \hat{\mathbf{u}}_{k|k}^{(N)}$, $\hat{\mathbf{U}}_{k|k} = \hat{\mathbf{U}}_{k|k}^{(N)}$.

Outputs: $\hat{\mathbf{x}}_{k|k}$, $\mathbf{P}_{k|k}$, $\hat{\mathbf{u}}_{k|k}$, $\hat{\mathbf{U}}_{k|k}$.

Different from the classical VB-EKF algorithm given in Ref. 12, the proposed algorithm uses a more accurate approximation of the nonlinear measurement function $\mathbf{h}(\mathbf{x}_k)$ obtained by performing linearization with an intermediate state estimate $\hat{\mathbf{x}}_{k|k}^{(i)}$,²¹ i.e.,

$$\mathbf{h}(\mathbf{x}_k) = \mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i)}) + \mathbf{H}_k^{(i)}(\mathbf{x}_k - \hat{\mathbf{x}}_{k|k}^{(i)}). \quad (5)$$

Then, an iteratively updated Jacobian matrix of the measurement function $\mathbf{H}_k^{(i)}$ is carried out to replace the fixed Jacobian matrix of the measurement function in the algorithm given by Ref. 12. In Algorithm 1, $N(\cdot; \mu, \Sigma)$ denotes the Gaussian distribution with mean μ and covariance matrix Σ . $IW(\cdot; \lambda, \Psi)$ denotes the inverse Wishart distribution with degree of freedom (dof) λ and inverse scale matrix Ψ . The iteratively updated Jacobian matrix of the measurement function $\mathbf{H}_k^{(i)}$ is expressed as

$$\mathbf{H}_k^{(i)} = \begin{bmatrix} h'_1, h'_2, \dots, h'_p \\ 0, 0, \dots, 0 \end{bmatrix}^T, \quad (6)$$

where h'_p ($p = 1, 2, \dots, P$) is given by¹³

$$h'_p = -2\pi f \frac{r}{c} \sin\left(\hat{\theta}_{k|k}^{(i)} - 2\pi \frac{p}{P}\right) \text{real}\left(jA_p\left(\hat{\theta}_{k|k}^{(i)}\right)s(k)\right), \quad (7)$$

where $s(k)$ is the true target signal at the k th time step. Since the true signal of the target is unknown in engineering applications, the estimate of the signal $\hat{s}_k$ is utilized instead of the real one. $\hat{s}_k$ can be obtained by using the maximum likelihood approach as²²

$$\hat{s}_k^{(i)} = \left(\mathbf{A}^H(\theta_{k|k}^{(i)})\mathbf{A}(\theta_{k|k}^{(i)})\right)^{-1}\mathbf{A}^H(\theta_{k|k}^{(i)})\text{Hilbert}(z(k)), \quad (8)$$

where $\mathbf{A}(\theta_{k|k}^{(i)})$ is the array manifold vector given by Eq. (4), $(\cdot)^H$ and $\text{Hilbert}(\cdot)$ denote the conjugate transposition and the Hilbert transform, respectively. It is easy to recognize that step 5 of the algorithm for calculation of $\mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i)})$ and $\mathbf{H}_k^{(i)}$ is unnecessary to operate after the first iteration to reduce computing load.

4. Experimental results

4.1 Experimental scenario

The data from an acoustic experiment in the South China Sea in July 2021 are used to verify the superiority of the proposed VB-EKF. An acoustic emission transducer is placed 3 m under the sea surface. A ship equipped with the acoustic signal emission system and GPS keeps moving when an acoustic signal at 170 Hz is emitted continuously for 7500 s with a source level of 145 dB.

An underwater buoy system shown in Fig. 2(a) is placed 20 m under the sea surface before the experiment. A uniform circular hydrophone array with a radius of 1 m and 12 elements is fixed on the underwater buoy along with a digital acquisition system and GPS. During the acoustic signal emitting, the digital acquisition system sampled the output of the array at a frequency of 8192 Hz. Before the experiment, the sound velocity is measured to be constant at 1544 m/s at a depth from 0 to 20 m with a CTD profiler. The GPSs record the locations of the acoustic signal emission system and the underwater buoy system every second. Considering the bearing angle of the target is measured clockwise with respect to the north, the bearing angle trajectory is calculated by the GPS data.

The tracking scenario is shown in Fig. 2(b), and the longest range between the target and the sonar array is 5462 m and the shortest range is 3113 m. In addition, the total distance that the target moved during the tracking

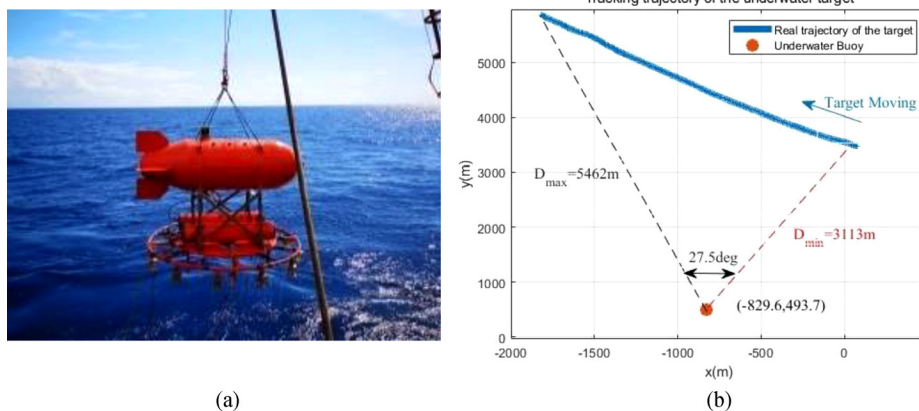


Fig. 2. (a) Photo of the underwater buoy system. (b) Tracking scenario.

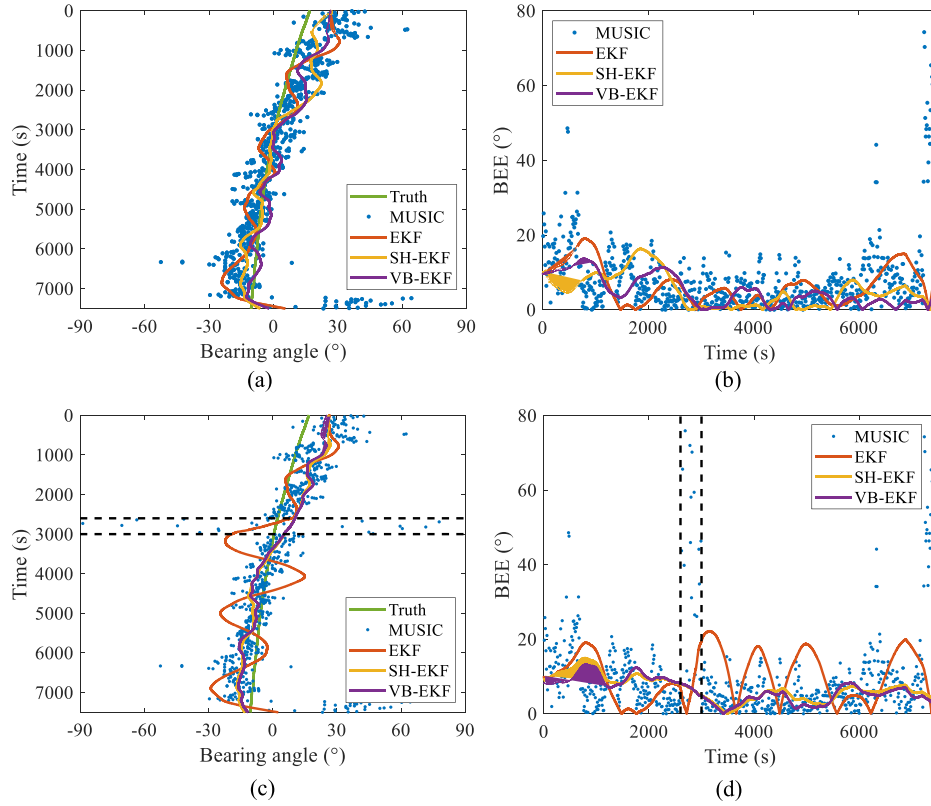


Fig. 3. Experimental results of MUSIC, EKF, SH-EKF, and VB-EKF. (a) Trajectories obtained using raw data, (b) BEEs obtained using raw data, (c) Trajectories obtained using the data with added noise, (d) BEEs obtained using the data with added noise.

period is 3059 m with the total bearing angle change with 27.5 degree at an approximate constant angular velocity of 3.7×10^{-3} deg/s.

4.2 Verification of VB-EKF for DOA tracking

The proposed VB-EKF are tested using the experimental data of all 7,500 s. The multiple signal classification (MUSIC), which is a traditional DOA estimation, the EKF, and the SH-EKF for DOA tracking¹³ are also tested for comparison. The measurement interval is set to be 1 s. The scanning bearing angle of the MUSIC varies every 1° from 0° to 360° . The initial state estimates of the EKF, SH-EKF, and the VB-EKF are both set to $\hat{X}_0 = [\hat{\theta}_{CBF}^0, 0]$, where $\hat{\theta}_{CBF}^0$ is the bearing angle estimated by using MUSIC at $k=0$. In addition, we suppose the first step DOA can be obtained by an accurate DOA estimation algorithm (i.e., MVDR or MUSIC) and the angular velocity of the target is small, P_k is initialized to $P_0 = \text{diag}([1, 10^{-5}])$ to represent the initial guess errors. The covariance matrix of the process noise Q_k is set as 10^{-4} for the EKF, SH-EKF, and the VB-EKF according to the reference value given previously. The iteration number N of the Iteration measurement update procedure in Algorithm 1 is set as 10. The MNCM is set as I_{MP} for the EKF. The estimation of the MNCM $\hat{R}_k$ is initialized to I_{MP} , and the forgetting factor b is set to 0.995 for the SH-EKF. The parameters $\hat{u}_0$, $\hat{U}_0$, and ρ of the VB-EKF are set to 16, $16I_{MP}$, and 0.997, respectively.

To numerically compare the bearing angle estimation precision of the previous methods, the bearing angle estimation error (BEE) and the average BEE (ABEE) are defined as the performance metrics, which are given as follows:

$$BEE(k) = |\hat{\theta}_k - \theta_k^{GPS}|, \quad (9)$$

$$ABEE(k) = \sqrt{\frac{1}{K} \sum_{k=1}^K (\hat{\theta}_k - \theta_k^{GPS})^2}, \quad (10)$$

where $\hat{\theta}_k$ is the bearing angle estimate at time step k and θ_k^{GPS} is the true bearing angle obtained by the GPS data.

In addition, since the real covariance matrix of the measurement noise in the sea trial cannot be obtained accurately, the real covariance matrix of the environment noise is approximated by received data from the sonar array without

Table 1. ABEEs of MUSIC, EKF, SH-EKF, and VB-EKF.

Data	ABEE (°)			
	MUSIC	EKF	SH-EKF	VB-EKF
Raw data	13.1	8.2	7.1	6.3
With noise	22.2	12.2	7.3	6.5

the target's signal. The error of the estimated covariance matrix of the measurement noise and the true covariance matrix is defined as the Euclidean distance (ED) as follows:

$$ED = \frac{1}{P} \sqrt{\sum_{i=1}^P \sum_{j=1}^P (x_{est,ij} - x_{real,ij})^2}, \quad (11)$$

where $x_{est,ij}$ and $x_{real,ij}$ are the opponents in the i th row and j th column of the estimated covariance matrix and real covariance matrix of the measurement noise, respectively.

The DOA tracking trajectory obtained by the VB-EKF is shown in Fig. 3(a). The BEEs and the ABEEs are shown in Fig. 3(b) and Table 1, respectively. In addition, the one-step operating time for MUSIC, EKF, SH-EKF, and VB-EKF are shown in Table 2 to demonstrate the computational efficiency of the DOA tracking algorithms compared to the DOA estimation method. The results of the MUSIC, EKF, and the SH-EKF are also shown in Figs. 3(a) and 3(b) and Table 1 for comparison. Specifically, considering the complexity of the actual underwater environment, the MUSIC obtained bearing angles every 10 s by using measurements of 10 s for higher precision.

In addition, there are many factors in the ocean environment can make the environmental noise varies uncertainly.²³ For example, the turbulence in the form of irregular random flow in the ocean and the pressure inside the turbulence produce acoustic effects. The formation and extinction of the turbulence will result in the variation of the environmental noise. Also, the cavitation or bursting of bubbles and the turbulent pressure caused by sea waves and winds generate noise. Thus, the variation of sea conditions and wind strength results in variation in environmental noise. Meanwhile, the radiated noise from distant ships is also an important source of environmental noise. The variation of the position, sailing status, and number of ships will also cause variation in environmental noise. Such variation of environmental noise results in the variation of statistical characteristics of environmental noise, which will make the covariance matrix of the measurement noise shift during the DOA tracking period.

Thus, in order to further test the robustness of the proposed VB-EKF to uncertain measurement noise, a period of correlated Gaussian noise generated by a technique from Ref. 24 is added to the raw measurement data from 2600 s to 3000 s, and then the methods are tested again. The results are shown in Figs. 3(c), 3(d), and Table 1. Also, the Euclidean distance between the estimated covariance matrix and the real covariance matrix of the measurement noise by different DOA tracking algorithms is shown in Fig. 4.

Figures 3(a) and 3(b) show that the trajectories obtained by all methods using raw experimental data fluctuated around the true trajectory due to the effect of the underwater environmental noise. Since the MUSIC only depends on the batch measurements and does not utilize the target's kinematic information, the trajectory obtained by MUSIC is the most fluctuant and provided a large error when the raw measurements are in low SNR. The precision of the EKF is higher than that of the MUSIC since the EKF is based on the DOA tracking principle that not only utilizes the current measurement but also considers the kinematic model of the underwater target. The trajectory obtained by the robust DOA tracking algorithms, namely, the SH-EKF and the VB-EKF are closest to the true trajectory. The reason is that the robust DOA tracking algorithms provide more accurate Kalman filter gains than EKF due to the high-precision estimation of the MNCM, which can significantly improve the precision of the DOA tracking. Also, it can be found that the proposed VB-EKF has higher DOA tracking accuracy compared to the SH-EKF since the VB-EKF is analytically derived, which makes it a precise mathematical model of the unknown MNCM and makes it more accurately estimated. However, the data-inspiring SH-EKF outputs DOA tracking results with lower accuracy since the precision of the experimental data is not accurate enough and

Table 2. One-step computational time for MUSIC, EKF, SH-EKF, and VB-EKF.

Data	Run time (ms)			
	MUSIC	EKF	SH-EKF	VB-EKF
Raw data	1.17	0.11	0.13	0.70
With noise	1.21	0.11	0.14	0.75

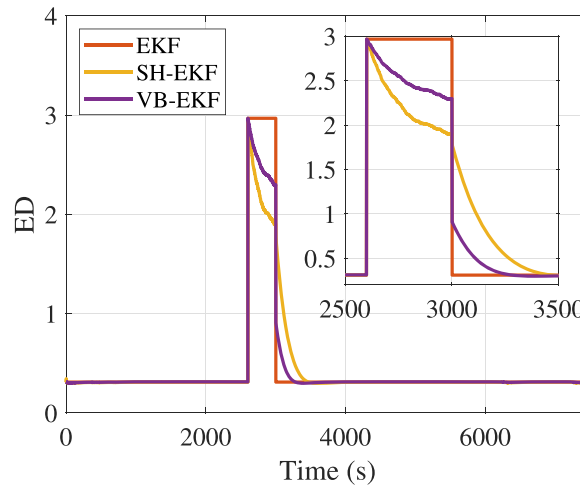


Fig. 4. ED between the estimated covariance matrix and real covariance matrix of the measurement noise.

the “good historical estimations” cannot be provided sufficiently to make the current DOA tracking results as accurate as the VB-EKF. Table 1 shows the high precision and robustness of the VB-EKF.

The results obtained by using the data with added noise given in Figs. 3(c), 3(d), and Table 1 show a similar conclusion to the one mentioned previously. Under this circumstance, the MUSIC, which only relies on the measurements, cannot deal with the noise shifting situation due to its ignorance of the target’s kinematics as prior information. In addition, without robust covariance matrix of the measurement matrix estimating techniques, the DOA tracking method baseline, namely, the EKF, has big fluctuations when the measurement noise shifting occurs. Nevertheless, the SH-EKF and VB-EKF can still provide acceptable DOA tracking results under this circumstance. However, the data-inspiring SH-EKF can only provide DOA tracking results with lower accuracy since the shifts of the measurement noise and the quality of the measurements cannot guarantee a sufficient tracking window of good historical estimations. On the contrary, the analytically derived VB-EKF can output more reliable tracking results as shown in Table 1. Also, as noted in Table 2, since the DOA tracking methods are based on the recursive estimation frame, the computational time of the EKF, SH-EKF, and VB-EKF costs much less computational time compared to the DOA estimation method (MUSIC), which only deals with batch measurements. In addition, the computational load of the VB-EKF is heavier than the EKF and SH-EKF since the VB-EKF has an iteration process. In the real tracking scenario, the tracking accuracy and computational efficiency can be balanced by engineering experience.

The estimation results of the covariance matrix of the measurement noise are shown in Fig. 4 by ED described by Eq. (11). During the measurement noise shifting period, both the SH-EKF and VB-EKF have the ability to make estimations of the covariance matrix of the measurement noise when compared to the EKF, which has a fixed one. By estimating the covariance matrix of the measurement noise, the SH-EKF and VB-EKF can make more reliable Kalman gain that can then deal with the shifting measurement noise. In addition, since the SH-EKF mainly depends on historically good estimations to output the current DOA tracking result, it can make faster covariance matrix estimations at the beginning of the shifting of the measurement noise. However, when the measurement noise becomes normal, affected by the historical estimations, the SH-EKF has fluctuations and finally leads to a lower estimation accuracy than the VB-EKF. On the contrary, since the VB-EKF is based on the analytical VB derivations, it has a slower covariance matrix estimating speed during the whole measurement noise shifting period. However, when the measurement noise becomes normal, the VB-EKF can still provide reliable estimations that finally make the VB-EKF have the highest DOA tracking accuracy.

5. Conclusion

A VB-EKF robust DOA tracking method is proposed and verified. The theoretical derivations have been given comprehensively and the performance has been proven by the real sea trial data. The experiment results show the proposed VB-EKF has more accurate estimation results and is also robust to uncertain underwater disturbance.

Future work should focus on enhancing the computational efficiency of the proposed VB-EKF since the parameter iteration steps during the VB procedure have a huge computational load, especially when the measurements are highly nonlinear. As a result, to make the proposed robust DOA tracking algorithm suitable for real time on-board utilization, enhancing its efficiency is necessary.

Also, the proposed VB-EKF still assumes the uncertain measurement noise to be Gaussian. However, non-Gaussian measurement noise is more common in real DOA tracking scenarios. To make the robust DOA tracking technique more accurate, nonlinear and non-Gaussian tracking schemes should be considered to work with the VB technique.

Acknowledgment

This work was supported by the National Natural Science Foundation of China under Grant No. 12104113 and the foundation of the State Key Laboratory of AFDL under Grant No. 6142210200103.

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